

Comparing Exploration Algorithms on Ms. Pac-Man

A Course-Project Experiment Design

Challenging Problems in Reinforcement Learning (SS 2026)

Topic: Exploration vs. Exploitation

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Abstract

We compare six exploration mechanisms—classical action-selection rules and deep intrinsic-motivation methods, including Learning Progress Monitoring (LPM)—on the Atari MS. PAC-MAN environment, measure their relative strength, and analyse *why* they differ. The study is built on the LPM codebase of Hou, An & Du (2026), *Beyond Noisy-TVs: Noise-Robust Exploration via Learning Progress Monitoring*. Two questions drive the design: **(RQ1)** how robust is each method to an injected noisy-TV, and **(RQ2)** does LPM’s advantage on *clean* Ms. Pac-Man grow with environment stochasticity—a probe of the hypothesis that the game’s random ghost movement already acts as aleatoric noise. All methods share one PPO base so that observed differences are attributable to the exploration mechanism rather than the base learner.

1 Motivation and research questions

The course topic is the exploration–exploitation trade-off. Classical exploration (ϵ -greedy, softmax/Boltzmann, UCB, Thompson sampling) was designed for bandit and value-based settings; deep RL instead uses intrinsic-motivation bonuses (prediction error, novelty, learning progress). The *noisy-TV* problem is the canonical failure mode of prediction-error exploration: an unlearnable source of randomness produces perpetually high prediction error, so a curiosity-driven agent fixates on it. LPM rewards the *improvement* of its dynamics model rather than the error level, and is therefore noise-robust by construction.

RQ1 (noise robustness, reproduction).

Under an action-triggered noisy-TV, which methods collapse and which survive? How large is LPM’s advantage over prediction-error curiosity (RND, ICM) and over a method explicitly built to filter aleatoric noise (AMA)?

RQ2 (stochasticity attribution, novelty).

LPM appears to win even on *clean* Ms. Pac-Man. Hypothesis: the game’s random ghost movement acts as aleatoric noise that fools prediction-error methods. Because the ghost AI is fixed in the ROM and cannot be tuned, we probe the *general* claim—“LPM’s advantage grows with environment stochasticity”—using ALE sticky actions (`repeat_action_probability`) as a controllable stochasticity knob.

2 Methods

All six trained methods use the same PPO base. Intrinsic-motivation methods add an intrinsic reward with weight β ; the per-step training reward is

$$r_t = r_t^e + \beta r_t^i, \quad (1)$$

with $\beta = 1$ throughout. The two classical baselines run on the same PPO base: plain PPO is the entropy-regularised stochastic policy (i.e. softmax/Boltzmann exploration), and ε -greedy adds a uniform-random action layer with probability $\varepsilon = 0.1$.

Class	Method	Mechanism (intrinsic reward r^i or rule)
Learning progress	LPM	improvement of the dynamics model, $\mathbb{E}[\varepsilon^{(\tau-1)} - \varepsilon^{(\tau)}]$
Prediction error	RND	error vs. a fixed random target network
Prediction error (feat.)	ICM	inverse-dynamics feature prediction error
Noise-robust curiosity	AMA	curiosity with aleatoric-uncertainty filtering
Classical (softmax)	PPO	entropy-regularised stochastic policy (no r^i)
Classical (ε -greedy)	PPO + ε	random action w.p. $\varepsilon = 0.1$

Table 1: The six trained methods. UCB and Thompson sampling are *discussion-only*: neither has a scalable deep-RL drop-in (UCB needs visit counts/uncertainty; Thompson needs posterior sampling), which itself illustrates why count-based bonuses and LPM are their practical descendants.

3 Experimental design

Two manipulation axes are kept separate to bound the run count.

RQ1 — noise axis. Each method is run in the *clean* game and the *noisy* variant (action-noise: two idle actions replace the observation with a random CIFAR-10 image), $\times 3$ seeds: $6 \times 2 \times 3 = 36$ runs.

RQ2 — stochasticity sweep (clean only). ALE sticky-action probability $p \in \{0.0, 0.25, 0.5\}$; the $p = 0.0$ point reuses the RQ1 clean runs. The sweep covers the four most diagnostic methods (LPM, RND, ICM, plain PPO) $\times \{0.25, 0.5\} \times 3$ seeds = 24 runs.

Budget. ≈ 60 runs $\times \sim 20$ min on Apple-Silicon MPS ≈ 20 h. Each run trains 10^6 aggregate frames, 16 parallel environments, PPO with learning rate 10^{-4} , clip 0.1, 3 epochs, 8 minibatches, $\gamma = 0.99$, GAE.

4 Metrics

Let S denote the final extrinsic game score (the mean episode return over the last 10% of updates), reported as mean $\pm$ std over seeds.

- **RQ1 headline — noise-robustness drop:**

$$\text{Drop} = \frac{S_{\text{clean}} - S_{\text{noisy}}}{S_{\text{clean}}}. \quad (2)$$

Prediction: LPM (and AMA) smallest; RND/ICM largest. Reproduces the paper’s central result.

- **RQ2 headline — stochasticity-advantage curve:** $S_{\text{LPM}}(p) - S_{\text{RND}}(p)$ and the same vs. ICM, as functions of the sticky probability p . Prediction: increasing $\Rightarrow$ supports the hypothesis.
- **Attribution (auxiliary):** learning curves (S vs. frames), and the per-method intrinsic-reward trajectory. LPM’s r^i should decay toward 0 as the world model converges; RND/ICM under noise should stay elevated—evidence of being “stuck on noise.” Intrinsic-reward magnitudes are *not* comparable across methods (different scales), so only within-method trends are interpreted.

5 Infrastructure and reproducibility

The embedded LPM Atari snapshot did not run out of the box; eight import/shape/correctness fixes were required (channel-first frames, ALE registration in subprocess workers, enabling the intrinsic reward, removing a Sigmoid decoder incompatible with normalised targets, MPS device support; all logged in `LPM_exploration/UPSTREAM.md`). Each run writes a per-update CSV (`update`, `frames`, `fps`, `ep_score_mean`, `ep_score_std`, `ep_len_mean`, `int_rew`, `ext_rew`, `pred_loss`, `unc_loss`, `dist_entropy`, `value_loss`, `action_loss`). A grid runner launches all runs sequentially on MPS (resumable: completed CSVs are skipped); an analysis script aggregates the CSVs into the results table and three figures.

6 Limitations (stated up front)

1. **Sticky actions $\neq$ ghost randomness.** Sticky actions are a controllable *proxy* for aleatoric stochasticity, not an isolation of ghost behaviour. RQ2 therefore tests the general “stochasticity $\rightarrow$ LPM edge” claim, with ghost randomness as qualitative motivation only.
2. **Short budget, small sample.** 10^6 frames $\times$ 3 seeds yields *qualitative trends*, not rigorous statistical significance. (At this scale a human casually reaches ~ 7000 while these agents reach ~ 500 ; maximising score is not the goal.)
3. **No per-state localisation.** We log aggregate intrinsic reward, not its distribution over game states, so the noisy-TV mechanism is shown indirectly.
4. **Episodic-bonus family absent.** EME and TDD/EDT model files are missing from the snapshot, so the comparison spans the curiosity, learning-progress, and classical families, not episodic bonuses.